# Integrity Concerns -- Guided Notes

Unit tp-6.2 | CompTIA Network+ N10-009 Obj. FC0-U71 6.1 | N10-008 FC0-U71 6.1

Name: ___________________________________ Date: ____________ Period: ______

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*Complete each question using the lesson reading. Write in the blank or answer in the space provided.*

### Question 1

Integrity in security means that data is _______, _______, and has not been _______ without authorization. An integrity violation occurs when data is modified by someone who was not _______ to make that change -- whether by an attacker, a system error, or accidental editing.

### Question 2

Hashing is the primary technical control for verifying integrity. A hash function produces a fixed-length _______ (digest) from input data. If even _______ bit of the data changes, the hash output changes completely. To verify integrity, compare the _______ hash to the _______ hash after transmission or storage.

### Question 3

SHA-256 (Secure Hash Algorithm 256-bit) is a widely used cryptographic hash function. It is _______ -way -- you cannot reverse the hash to recover the original input. MD5 produces a 128-bit hash but is considered _______ for security purposes because collisions can be engineered. A collision is when two _______ inputs produce the _______ hash output.

### Question 4

A digital _______ ensures both integrity and _______ of a message or document. The sender hashes the message and _______ the hash with their private key. The receiver decrypts the hash with the sender's _______ key and re-hashes the message -- if the hashes match, the message is unmodified and the sender is authenticated.

### Question 5

Threats to integrity include: _______ injection (attackers insert malicious data into a system), _______ -in-the-middle attacks (attacker modifies data during transmission), and _______ (attackers redirect users to fraudulent websites by corrupting DNS records). All three result in users receiving _______ data.

### Question 6

File integrity monitoring (FIM) tools alert administrators when critical system files are _______ modified. They do this by storing _______ of key files at baseline and comparing regularly. Examples include AIDE (Linux) and Tripwire. FIM is especially important on _______ and _______ servers where unauthorized changes could indicate a compromise.

### Question 7

_______ controls like audit logs and version control help maintain and demonstrate integrity. A _______ trail records every change to data including who made it, when, and what changed. This is essential for _______ investigations and regulatory _______ (HIPAA, PCI-DSS require audit logging).

### Question 8 -- Real World Application

REAL WORLD: A bank publishes software update files on their website so customers can download the latest mobile app. An attacker compromises the web server and replaces the legitimate installer with a malware-laced version. The file is the same size and has the same name. How would publishing a SHA-256 hash prevent customers from installing the malicious file?

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*Download the answer key from the class server:*

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wget http://[SERVER]/resources/network-engineer/tp-6-2-integrity-concerns/tp-6-2-integrity-concerns-answer-key.pdf
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